

Module 7

Advanced Git for Collaboration

Learning objectives

- Create and switch feature branches
- Merge branches and resolve conflicts
- Understand submodules for shared IP
- Use git blame and bisect conceptually

Prerequisites

- Module 6 (add, commit, push)

Learning path

Learning Path

Diagram: Learning Path

(render with mmdc when available)

flowchart LR

A[Branches] --> B[Merge]

B --> C[Conflicts]

C --> D[Submodules]

D --> E[blame & bisect]

E --> F[Module 8]

Course context

- Team labs use branches per feature
- Conflicts happen when two people edit same lines
- Verification IP may live in submodules

Core topics

Self-check

```
$ ./scripts/module7.sh --check
```

Terminal: module_check

```
#####  
Module 7: Environment self-check  
#####  
  
[OK] git available: git version 2.43.0  
[OK] module7 directory exists: /home/yongfu/proj/universal-verification-methodology/  
[OK] conflicts/sample_conflict.txt exists  
  
[OK] All required checks passed.
```

Feature branch

```
git checkout -b feature/lab2  
git merge main
```


Merge conflict format

```
$ head -15 module7/examples/conflicts/sample_conflict.txt
```

Terminal: conflict_sample

This file shows how Git marks merge conflicts (for reference only).
In a real conflict, Git inserts these markers into your file.

<<<<<< HEAD

This is the version from the current branch (e.g. main).
Keep this if you want the main-branch change.

=====

This is the version from the branch being merged (e.g. feature/x).
Keep this if you want the feature-branch change.

>>>>>> feature/x

To resolve: choose one version (or combine both), then delete the markers and the line you don't

Branches

```
$ git help branch 2>/dev/null | head -8
```

Terminal: git_branch_help

GIT-BRANCH(1)

Git Manual

NAME

git-branch - List, create, or delete branches

SYNOPSIS

git branch [--color[=<when>] | --no-color] [--show-current]
[-v [--abbrev=<n> | --no-abbrev]]

Practice

Exercises

- Create branch in git_demo
- Resolve sample_conflict.txt on paper
- Try git submodule update --init

Summary & next steps

- Branching and merging
- Conflicts and submodules
- Next: Module 8 — end-to-end workflow
- Next: Module 8: End-to-End Project Workflow